

Details: In this assignment, your goal is to design and train neural network models to tackle a Chunking/shallow parsing task. Specifically, for a given dataset, your objective is to develop Recurrent and Transformer models, train them effectively, and analyze their performance. Finally, you will report your findings based on the results obtained.

### Marking Rubric for CA3

| Criteria                        | Details                                                                                                                          | Points |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------|
| Code: Protocol                  | Clear definition of the data split / protocol                                                                                    | 5      |
| Code: DataLoader                | Appropriate dataloader with correct encoding, vocabulary, etc                                                                    | 13     |
| Code: Networks and Components   | Correct definition of network architectures as well as necessary components                                                      | 20     |
| Code: Train                     | Train the models and plot graphs                                                                                                 | 17     |
| Code: Test                      | Test using the appropriate set                                                                                                   | 5      |
| Report: Protocol                | Clearly and briefly define your experimental protocol                                                                            | 5      |
| Report: DataLoader              | Explain the main details of your dataloader, including any pre-processing performed (such as encoding, vocabulary creation, etc) | 12     |
| Report: Networks and Components | Briefly describe the exploited network architectures, as well as the components required for training                            | 13     |
| Report: Results                 | Report and discuss the table with the training results - clearly define the best model                                           | 5      |
| Report: Test                    | Report and discuss a table and confusion matrix with the final results using the best model on the appropriate set               | 5      |